



Microstructure and mechanical properties of multi-component AlCrFeNiMo_x high-entropy alloys



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ABSTRACT

The AlCrFeNiMo_x ($x = 0, 0.2, 0.5, 0.8, 1.0$) high entropy alloys were produced by vacuum arc melting, and their microstructure and mechanical properties were investigated. AlCrFeNi, AlCrFeNiMo_{0.2}, and AlCrFeNiMo_{0.5} alloys contained two body centered cubic (BCC) phases which were AlNi-type intermetallic compound and FeCr-type solid solution. The diffraction peaks of the two BCC phases separated with the increase of Mo content while Mo element preferred to dissolve into FeCr-type solid solution. When the value of x was higher than 0.5, the FeCrMo-type σ phase appeared and replaced the FeCr-type solid solution. The AlCrFeNiMo_{0.2} had the highest fracture strength of 3222 MPa and plastic strain of 0.287. The hardness increased obviously from HV472.4 to HV911.5 with the addition of Mo element. The solid-solution strengthening of the BCC matrix and the σ phase hardening were the two main factors that strengthened the alloys. The AlCrFeNiMo_{0.2} high entropy alloy possesses the excellent mechanical properties which imply its potential application in industrial areas.

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1. Introduction

Conventional alloys include only one principal element and their performance is improved with minor additions of other elements, such as iron, aluminum, magnesium, and copper. In order to break the bottleneck stage of traditional alloys design concept, a new alloy system called high entropy alloys (HEAs) has been proposed by Yeh et al. for the first time [1,2]. HEAs generally include at least five principal elements with the concentrations of each element being between 5 and 35 at.% [2,3]. Later, Zhang et al. proposed the solid solution formation rules [4–6] and perfected the theory of HEAs. The previous studies have shown that HEAs only form the simple face center cubic (FCC), body center cubic (BCC) or FCC + BCC solid solution structures instead of lots of intermetallics, which can be explained using the high mixing entropy theory [7–10]. Besides, HEAs are advantageous in some excellent properties, such as high hardness [11,12], high elevated-temperature strength [13,14], high corrosion resistance [15,16], high wear resistance [17,18] and so on. These advantages imply that HEAs have a wide potential application in wear-resistance tools, molds, furnace and structural components.

Among the previous HEAs investigated, Al–Co–Cr–Cu–Fe–Ni alloy system has an outstanding mechanical performance [2,4]. However, the element Cu which is easily segregated in the interdendritics will deteriorate its mechanical properties. Therefore,

the research focused on the Al–Co–Cr–Fe–Ni alloy system instead. Then the microstructure and properties of AlCoCrFeNiMo_x [19], AlCoCrFeNiTi_x [10], AlCoCrFeNiC_x [20], and AlCoCrFeNiSi_x [21] alloys begin to report one after another. In this paper, element Cu and Co elements are removed and the AlCrFeNi alloy is selected as the matrix alloy. In accordance with the HEAs design concept, Mo element is added to the AlCrFeNi alloy and the variation of its microstructure and properties are investigated.

2. Experimental

The purity of each raw materials used in this study is shown in Table 1. The AlCrFeNiMo_x (when $x = 0, 0.2, 0.5, 0.8, 1.0$, the alloys were denoted as Mo00, Mo02, Mo05, Mo08 and Mo10, respectively) alloys were prepared by arc melting mixture of pure metals in a Ti-gettered high purity argon atmosphere. After the vacuum pressure reached to 4×10^{-3} Pa, melting and casting were performed with the pressure of 0.01 atm by backfilled to argon (Ar). The alloys were repeatedly melted with electromagnetic stirring at least five times for homogeneity. Then the alloys were directly solidified in a water-cooled copper hearth and the solidified ingots were about 30 mm in diameter and 15 mm in thickness (approximately 70 g). The samples were then polished and etched in diluted aqua regia before examining microstructure by a scanning electron microscope (SEM, Zeiss supra 55) operated at secondary electron mode and with an accelerating voltage of 15 kV. Chemical compositions of different phases were analyzed by SEM energy dispersive spectrometry (EDS), and the crystal structure of as-cast samples was characterized by an X-ray diffractometer (XRD, EMPYREAN with Cu radiation target) with the 2θ scan ranging from 20° to 100° at a speed of $4^\circ/\text{min}$. For the compressive test, specimens with a diameter of 5 mm and a length of 10 mm were prepared from the master ingot by electro-spark wire-electrode cutting and the test was carried out at a speed of 3 mm/min. The hardness measurement was conducted using a Vickers hardness tester (MH-6L) under a load of 1000 g for 15 s. Five measurements were made on each sample to obtain the averaged experimental date.

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Table 1

Raw materials and their purity (mass fraction).

Component	Al	Cr	Mo	Fe	Ni
Purity (%)	99.9	99.7	99.9	99.9	99.9

3. Results and discussion

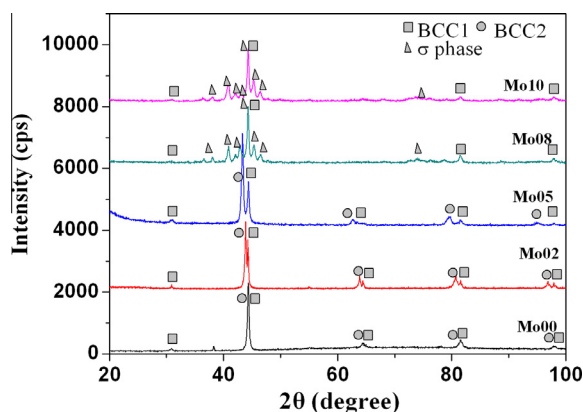
3.1. Crystal structure

The XRD patterns of the AlCrFeNiMo_x alloys are shown in Fig. 1. Mo00, Mo02 and Mo05 alloys exhibit two BCC phases. In the Mo00 alloy, the diffraction peaks of the two BCC phases (denoted as BCC1 and BCC2, respectively) coincide with each other due to the similar lattice parameters of the two phases [22]. However, with the addition of Mo element into the Mo00 alloy, the lattice parameters of the two BCC phases change differently, resulting in the separation of their diffraction peaks. The (1 1 0) peak of the BCC2 phase shifts to the lower 2θ with the Mo content increase from x = 0 to x = 0.5, as shown in Fig. 1. This shift is due to the variation of lattice parameters of the BCC2 phase, which are estimated to be 0.2885, 0.2914, and 0.2951 nm from the strongest (1 1 0) peak using the Bragg equation. For Mo08 alloy and Mo10 alloy, besides BCC1 diffraction peaks, the diffraction peaks of σ phase, which can be identified as FeCrMo-type phase by XRD, will also appear with the lattice parameters of a = 0.917 nm and c = 0.474 nm. The transition from BCC2 phase to σ phase indicates that the solution limit of the Mo element into the BCC2 phase should be less than 28.4 at.% which corresponds to x = 0.8.

According to the Gibbs phase rule: $F = C - P + 1$ (F : degree of freedom, C : number of components, P : number of phases), the maximum number of equilibrium phases at constant pressure is $P = C + 1$. In this paper, the total number of phases is below the maximum equilibrium number allowed by the Gibbs phase rule, which might be due to the high mixing entropy effect. However, according to the Hume-Ruthery rule, the small atomic size difference, the similar electro-negativity and the same lattice structure are the key factors in solid solution formation. Accordingly, for the alloy series, the atomic-size difference (ΔR), the mixing entropy (ΔS), the mixing enthalpy (ΔH), the electro-negativity difference (ΔX), and valence electron concentration (VEC) of the AlCrFeNiMo_x alloys are defined and calculated as [4–6]:

$$\Delta R = 100 \sqrt{\sum_{i=1}^N C_i \left(1 - \frac{R_i}{\bar{R}}\right)^2} \quad (1)$$

$$\Delta S = -R \sum_{i=1}^N (C_i \ln C_i) \quad (2)$$

**Fig. 1.** XRD patterns of the AlCrFeNiMo_x alloys.

$$\Delta X = \sqrt{\sum_{i=1}^N C_i (X_i - \bar{X})^2} \quad (3)$$

$$\Delta H = \sum_{i=1, i \neq j}^N \Omega_{ij} C_i C_j \quad (4)$$

$$\text{VEC} = \sum_{i=1}^N C_i (\text{VEC})_i \quad (5)$$

where N is the number of the components in an alloy system, C_i and C_j are the atomic percentages of the i th and j th element, R_i is the atomic radius of i th element which can be obtained from Table 2. $\bar{R} (= \sum_{i=1}^N C_i R_i)$ is the averaged atomic radius, $\Omega_{ij} (= 4\Delta H_{ij}^{\text{mix}} C_i C_j)$ is the regular melt-interaction parameter between i th and j th elements, $\Delta H_{ij}^{\text{mix}}$ is the mixing enthalpy of binary AB alloys which has been listed in Table 3. X_i is the Pauling electronegativity for the i th element. $\bar{X} (= \sum_{i=1}^N C_i X_i)$ is the averaged electronegativity. $(\text{VEC})_i$ is the VEC for individual element. X_i and $(\text{VEC})_i$ can be obtained from Table 2. ΔS , ΔH , ΔR , ΔX , and VEC for AlCrFeNiMo_x alloys are all listed in Table 4. Obviously, with the increase of Mo content, electro-negativity, mixing entropy, and mixing enthalpy increase while atomic size difference firstly increases and then decreases. The FeCrMo-type σ phase replaces the BCC2 phase with the increase of Mo content from x = 0.5 to x = 0.8, which indicates that the increase of the electro-negativity has stronger effect than combinational effect of the increased mixing entropy, mixing enthalpy, and the decreased atomic radius difference. Guo and Liu et al. have studied the effect of valence electron concentration on stability of FCC or BCC phase in high entropy alloys and came to the conclusion that BCC phases are more stable at lower VEC (<6.87) [5,23]. In this paper, the VEC of AlCrFeNiMo_x alloys, which decreases with the increase of Mo content, maintains to be less than 6.87. Mo00, Mo02, and Mo05 alloys all have stable BCC structure, which satisfies Guo's assumptions. However, the appearance of σ phase in Mo08 and Mo10 alloy cannot be explained solely by the VEC factor. Therefore,

Table 2

Properties of the experimental elements.

Experimental elements	Al	Cr	Fe	Ni	Mo
Metallic atomic radius (pm)	143	128	126	124	139
Pauling's electronegativity	1.61	1.66	1.83	1.91	2.16
VEC	3	6	8	10	6

Table 3

Chemical mixing enthalpies of the atomic pairs among the alloying elements (kJ/mol).

	Al	Cr	Fe	Ni	Mo
Al	–	–	–	–	–
Cr	–10	–	–	–	–
Fe	–11	–1	–	–	–
Ni	–22	–7	–2	–	–
Mo	–5	0	–2	–7	–

Table 4Parameters of ΔS , ΔH , ΔR , and ΔX for AlCrFeNiMo_x alloys.

Alloys	ΔS (J/K mol)	ΔH (kJ/mol)	ΔR (%)	ΔX	VEC
Mo00	11.53	–13.25	5.755	0.1221	6.75
Mo02	12.57	–12.65	5.777	0.1474	6.71
Mo05	13.15	–11.85	5.779	0.1722	6.67
Mo08	13.35	–11.15	5.755	0.1884	6.63
Mo10	13.38	–10.72	5.73	0.1962	6.6

several factors, such as ΔS , ΔH , ΔR , ΔX , VEC, need to be taken into consideration when making predications on the stability of TCP phases in high entropy alloys.

3.2. Microstructure

Fig. 2a presents a typical regular lamellar eutectic for Mo00 alloy and Fig. 2b is the magnification of the closed rectangle area in Fig. 2a. The layers become thicker with the increase of the distance from the center of the grain. Wang [22] has concluded that the eutectic structure is a mixture of the AlNi-type intermetallic compound and the FeCr-type solid solution, which matches the result

obtained in this paper. Table 3 lists the chemical mixing enthalpies of the atomic pairs among the experiment elements [24]. The mixing enthalpies between aluminum and nickel, chromium, and iron are -22 , -10 , -11 kJ/mol, respectively. Hence, aluminum attracts nickel to form the AlNi-type phase while the rest of elements form the FeCr-type phase. The typical eutectic cell microstructure of Mo02 alloy is presented in Fig. 2c and d. Compared with Mo00 alloy, the lamellar AlNi-type phase near the grain boundary becomes a triangular and blocky structure. The chemical composition of different regions of AlCrFeNiMo_x alloys, analyzed by EDS, is summarized in Table 5. It should be noted that in Mo02 alloy, 1.6 at.% Mo element dissolves into the AlNi-type intermetallic compound

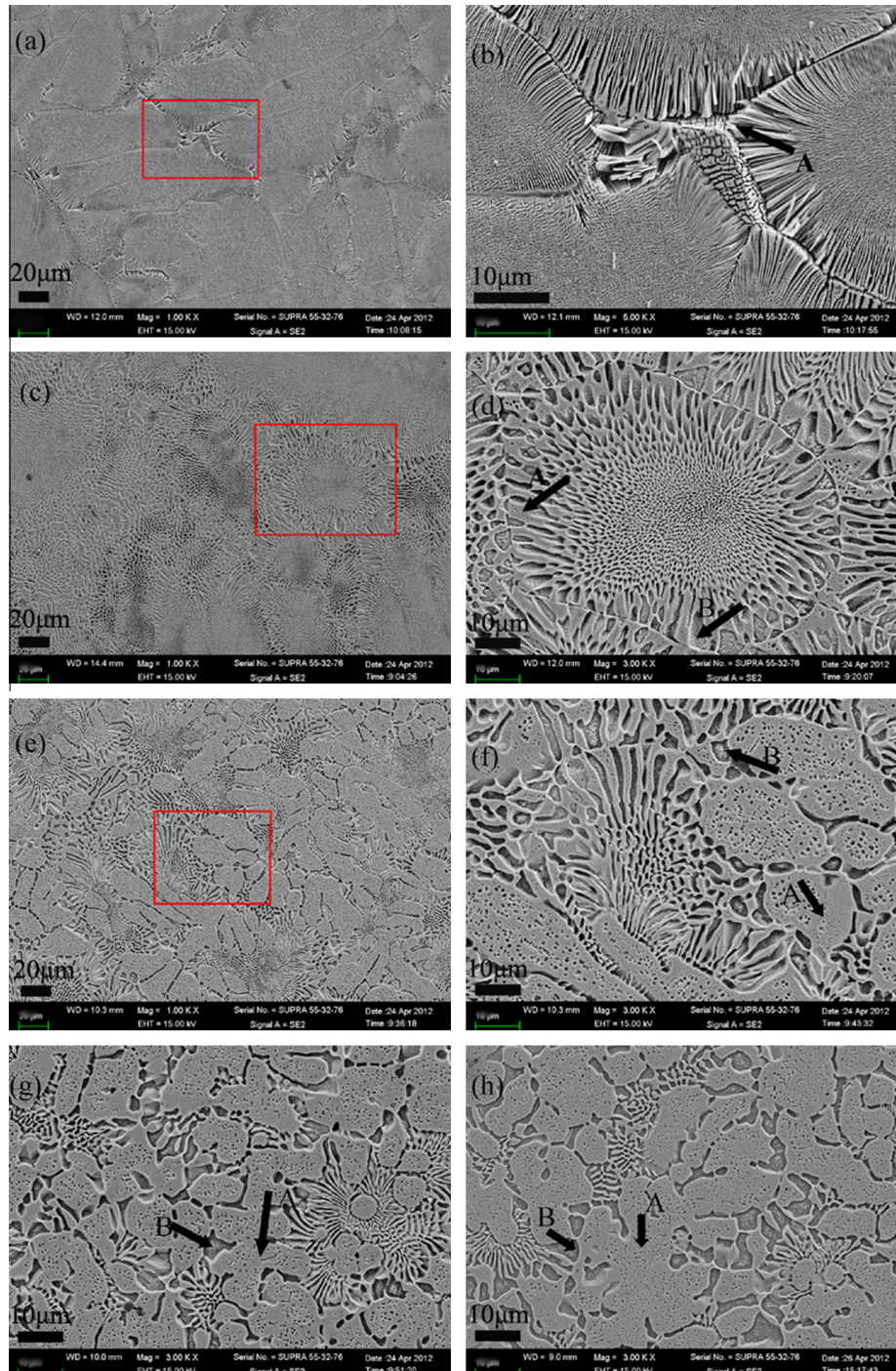


Fig. 2. SEM images of the AlCrFeNiMo_x alloys; (a) $x = 0$, low magnification; (b) $x = 0$, high magnification; (c) $x = 0.2$, low magnification; (d) $x = 0.2$, high magnification; (e) $x = 0.5$, low magnification; (f) $x = 0.5$, high magnification; (g) $x = 0.8$; (h) $x = 1.0$.

(mark as A) while 7.6 at.% dissolves into the FeCr-type solid solution phase (mark as B), which indicates that element Mo has much larger solubility in the FeCr-type solid solution than that in the AlNi-type intermetallic compound. With increased Mo content from $x = 0.2$ to $x = 0.5$, the microstructure exhibits a typical hypoeutectic structure wherein the primary phase is the FeCr-type solid solution (mark as A), and the lamellar eutectic structure is a mixture of the FeCr-type solid solution (mark as A) and AlNi-type intermetallic compound (mark as B), as can be seen in Fig. 2e and f. From Table 5, it can be seen that the up to 17.4 at.% Mo element dissolves into the FeCr-type phase. When the Mo content exceeds $x = 0.5$, the microstructure changes from hypoeutectic to hypereutectic structure. The primary phase is not the FeCr-type solid solution but the FeCrMo-type σ phase (mark as A), as shown in Fig. 2g and h. In addition, the EDS result shows that the primary phase contains Mo ranging from 26.7 at.% to 28.5 at.%, Fe ranging from 26.5 at.% to 28.9 at.%, and Cr ranging from 27.8 at.% to 28.1 at.% for Mo08 and Mo10 alloys. Therefore, the σ -type FeCrMo phase is further confirmed by XRD and EDS results. Tsai et al. observed a σ phase as the stable matrix phase [25]. In this paper, a FeCrMo-type σ phase as a stable matrix phase is also observed. Besides, it is rare that the FeCrMo-type σ phase appears as the primary phase and accounts for a large proportion in the high entropy alloy systems.

3.3. Mechanical properties

There exists a big difference in the mechanical properties of the same alloy with different cooling rates during solidification process. Zhang et al. has investigated the effect of the cooling rate and size effect on the microstructure and mechanical properties of AlCoCrFeNi high entropy alloy [26] and found that when the diameter of suction casting cylindrical rods changed from 2 mm to 10 mm, the compressive fracture strength decreased from 2920 MPa to 1830 MPa. In the industrial production, it is impossible to obtain such a high cooling rate for suction cast cylindrical rods with the diameter of 2 mm. To approach the actual conditions, the compressive samples ($\phi 5$ mm $\times$ 10 mm) in this paper were taken from the ingots directly solidified in the cold copper hearth.

Figs. 3 and 4 show the compressive engineering stress–strain (Mo10 alloy excluded) and the Vickers hardness curves of AlCrFeNiMo $_x$ alloys with different Mo contents, respectively. The yield strength σ_y , fracture strength σ_b , plastic strain ε_p and the value of Vickers hardness (HV) are listed in Table 6. As the Mo content increases from $x = 0$ to $x = 0.5$, the yield strength increases substantially from 1406.2 MPa to 1748.6 MPa. The alloy with a low content of Mo element about $x = 0.2$ has the highest fracture strength of 3222 MPa and plastic strain of 0.287. When the value of x increases from $x = 0.2$ to $x = 0.8$, the fracture strength and the plastic strain decrease significantly from 3222 MPa to 1512.5 MPa and from 0.287 to 0, respectively. On the contrary, the hardness value of the alloys increases significantly from Hv472.4 to Hv911.5.

Table 5

The EDS results of AlCrFeNiMo $_x$ alloys in different regions.

Alloys	Spectrum point	Al	Cr	Fe	Ni	Mo
Mo00	A	41.1	4.1	10.4	44.4	0
Mo02	A	14.6	29.4	34.4	14.0	7.6
	B	37.6	7.9	13.3	39.6	1.6
Mo05	A	8.6	31.8	34.4	7.8	17.4
	B	23.1	18.4	20.8	31.3	6.4
Mo08	A	10.1	28.1	28.9	6.2	26.7
	B	27.9	12.6	19.0	34.6	5.9
Mo10	A	8.9	27.8	26.5	8.3	28.5
	B	36.6	8.1	13.7	37.6	4.0

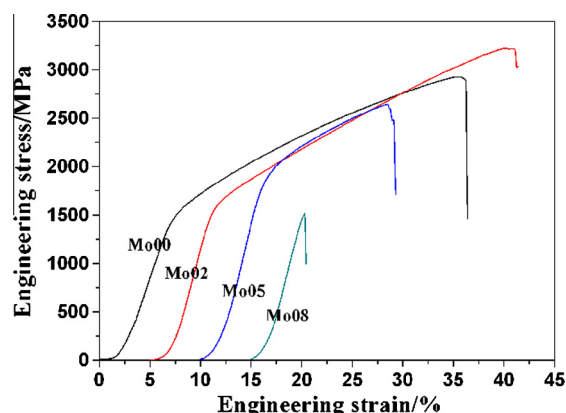


Fig. 3. Compressive engineering stress–strain curves of AlCrFeNiMo $_x$ alloys at room temperature.

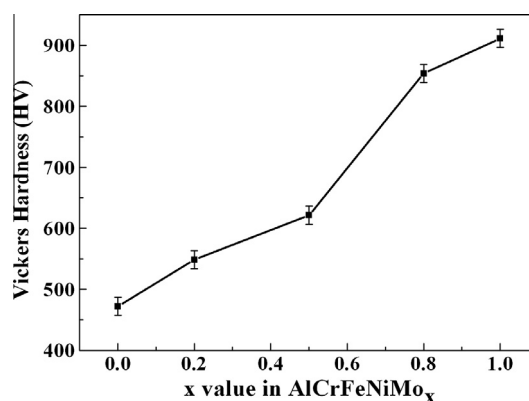


Fig. 4. Vickers hardness curves of AlCrFeNiMo $_x$ with different Mo contents.

Table 6

Room temperature mechanical test results for AlCrFeNiMo $_x$ alloys.

Alloy	σ_y (MPa)	σ_b (MPa)	ε_p (%)	HV
Mo00	1406.2	2926.5	28.5	472.4
Mo02	1487.4	3222	28.7	548.5
Mo05	1748.6	2643.5	12.6	621.5
Mo08	1512.5	1512.5	0	853.8

Compared with the suction casting ingots which solidified at a high cooling rate, it is rare that the Mo02 alloy which solidifies at a low cooling rate has such a high fracture strength and plastic strain. That is mainly related to the solid solution strengthening of Mo atoms. The plastic strain of Mo05 alloy drops substantially due to the microstructure transform from eutectic to hypoeutectic structure. And with a further increase of Mo element, the fracture strength decreases significantly due to the transformation from the FeCr-type solid solution to the FeCrMo-type σ phase. The FeCrMo-type σ phase with a tetragonal structure is a brittle Topological Close-Packed (TCP) phase. And it should be noted that the formation of TCP phase seriously damages the compression properties of the alloys. The increase in hardness appears to occur in three stages, firstly a gradual increase from $x = 0$ to $x = 0.5$, then a sharp increase from $x = 0.5$ to $x = 0.8$ followed by another gradual increase to the highest Mo content. The first increase is related to solid solution strengthening. From Table 3, it can be seen that Mo element has the second largest atomic radius and it can cause strong lattice distortion when dissolved into AlNi-type and FeCr-type phases. The fact is proved by the diffraction peaks of BCC2

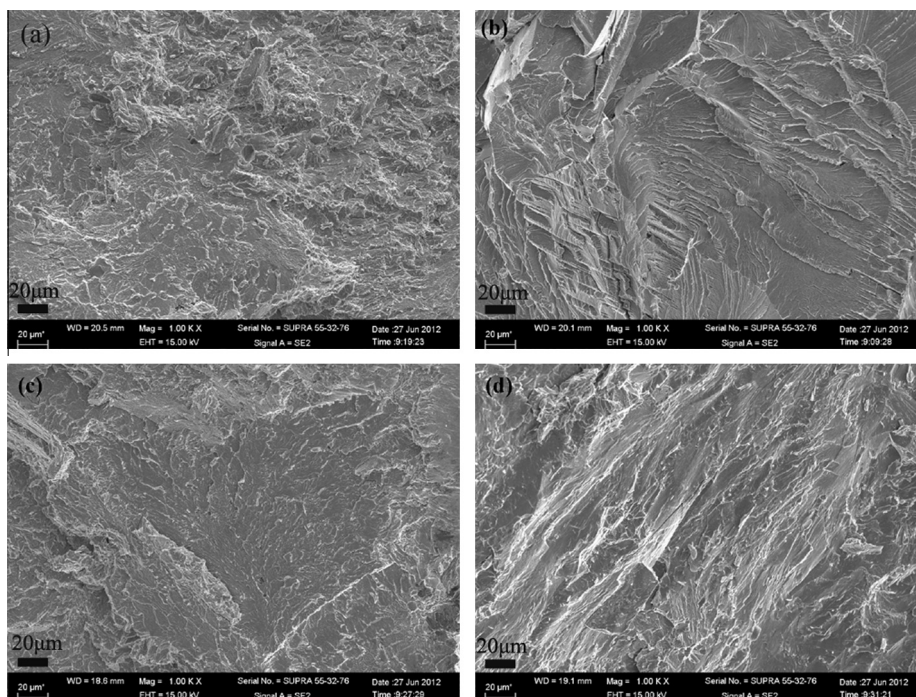


Fig. 5. SEM images of fracture surface of AlCrFeNiMo_x alloys; (a) $x = 0$; (b) $x = 0.2$; (c) $x = 0.5$; (d) $x = 0.8$.

phase shifting to the lower 2θ . The sharp increase in hardness beyond a Mo content of $x = 0.5$ is considered as the result of the FeCrMo-type σ phase. At room temperature the average hardness of σ phase is 1100–1300 kgf/mm² [27]. Therefore, the formation of FeCrMo-type σ phase is the most important factor of strengthening, which increases the hardness substantially. The increased volume fraction of the primary σ phase contributes to the last hardness increase stage in hardness as presented in Fig. 2h.

Fig. 5 shows SEM images of compression fracture surface of AlCrFeNiMo_x alloys. The fracture surface of torn edge, cleavage step and secondary crack of Mo00 alloy is shown in Fig. 5a and it can be seen that the fracture pattern is a typical quasi-cleavage fracture. For the Mo02 alloy, the river-like patterns and cleavage step formation in different parallel height alloys are presented in Fig. 5b, which is a visible trans-granular cleavage fracture pattern. Fig. 5c shows that in the Mo08 alloy, there are high-density short tear edges and cleavage steps, which are the characteristics of quasi-cleavage fracture. The bright and flat fracture surface shown in Fig. 5d indicates that the compression fracture pattern of Mo08 alloy is an absolutely brittle cleavage fracture.

4. Conclusions

In this study, the effects of the addition of various amounts of Mo on the microstructures and mechanical properties of AlCrFeNiMo_x ($x = 0, 0.2, 0.5, 0.8, 1.0$) high entropy alloys were investigated. XRD and SEM results revealed that the crystal structure transformed from two BCC phases to one BCC phase plus FeCrMo-type σ phase while the microstructure changed from eutectic structure to hypoeutectic structure and then to hypereutectic structure with the increase of Mo content. The two BCC phase involved are AlNi-type and FeCr-type phase. The hypoeutectic structure consists of the FeCr-type primary phase and a mixture of the FeCr-type solid solution and AlNi-type intermetallic compound lamellar eutectic structure while the hypereutectic structure consists of FeCrMo-type σ primary phase and a mixture of the FeCrMo-type σ phase and AlNi-type intermetallic compound lamellar eutectic structure.

It is rare that FeCrMo-type σ phase which replaced BCC2 phase entirely was observed as a primary phase in hypereutectic structure. The value of hardness (HV) increased from 472.4 to 911.5 due to the solution strengthening and the transformation from FeCr-type solid solution to FeCrMo-type σ phase. The compressive samples were obtained from master ingot ($\phi 30$ mm $\times$ 15 mm) which solidified at a relatively low cooling rate. With the increase of Mo contents, the fracture strength and the plastic strain first increased then decreased. Surprisingly, for Mo02 alloy, the fracture strength and plastic strain were as high as 3222 MPa and 0.287, which predicts its potential application in structural materials.

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